

Target-Population Fragility in Judge IV under Average Monotonicity

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Abstract

Judge-IV designs increasingly invoke average monotonicity when strong monotonicity is hard to defend. We show that average monotonicity does not preserve a stable target population: the response types receiving positive weight switch discretely when a judge’s treatment propensity crosses the assignment-weighted average. With heterogeneous treatment effects, the estimand remains a proper weighted average but may represent a different target population after an arbitrarily small design change. A standardized boundary-distance diagnostic flags fragile judges; in Philadelphia bail data, three of eight judges, covering a non-negligible share of cases, are fragile. Target-population stability is therefore an empirical object.

Keywords: Judge-IV designs; Average monotonicity; Target population; Complier weights; Pretrial detention

JEL Classification: C21, C26, C18

1 Introduction

Instrumental-variable designs based on the random assignment of judges have become a standard source of quasi-experimental variation, used to estimate the effects of incarceration, pretrial detention, disability insurance, and other consequential decisions.¹ Their interpretation rests on monotonicity. Strong monotonicity, the requirement that a judge

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¹Prominent applications include [Kling \(2006\)](#); [Maestas et al. \(2013\)](#); [Doyle Jr et al. \(2015\)](#); [Dobbie et al. \(2017, 2018\)](#).

who is stricter on average be weakly stricter in every case, is often hard to defend. Thus, a growing literature has turned to average monotonicity, a weaker condition that preserves the IV estimand as a proper weighted average of treatment effects even when judges rank cases differently (Frandsen et al., 2023; Chyn et al., 2025). That shift fixes the sign of the weights, but it leaves a separate question open. This paper asks whether the IV estimand under average monotonicity identifies the effect for a stable target population.

We answer this question in the smallest setting that displays the mechanism, a three-judge design with eight response types. Under strong monotonicity, the target population is defined by behavior, namely the individuals whose treatment changes when they face a stricter judge. Under average monotonicity, the set of positively weighted types depends instead on where each judge’s propensity lies relative to the assignment-weighted average $\bar{p}$. Two non-monotone types sit at that boundary with weights of opposite sign, so when a judge’s propensity crosses $\bar{p}$, one type leaves the positively weighted population and its mirror enters. If these two types have positive shares in their respective populations and distinct average treatment effects, the identified IV parameter need not be locally invariant, even though the propensity profile changed by an arbitrarily small amount. Average monotonicity continues to rule out negative weights, but it does not pin the estimand to a population fixed by behavior alone. Where Mogstad and Torgovitsky (2024) note that the target population identified under average monotonicity depends on the instrument distribution, we show that this dependence is discontinuous: an arbitrarily small change in the propensity profile can change which response types the estimand averages over, not only their weights. The same logic extends to any finite number of judges.

The boundary result has an empirical counterpart. Because researchers estimate judge propensities rather than observe them, we propose a standardized boundary distance that measures whether the sample places each judge securely on one side of $\bar{p}$, and we illustrate it in the Philadelphia bail data of Stevenson (2018). The exercise does not reject the design or test average monotonicity; it shows that target-population stability should be reported alongside balance, first-stage, and exclusion checks (Chyn et al., 2025; Goldsmith-Pinkham et al., 2025).

2 IV Weights under Average Monotonicity

We work in the standard judge-IV design with random assignment and exclusion, where the IV estimand is a weighted average of individual treatment effects. Let $Z_i \in \{1, \dots, J\}$ denote individual i ’s judge, with $\Pr(Z_i = z) = \lambda_z > 0$, and let $D_{iz} \in \{0, 1\}$ be the treatment individual i would receive if assigned to judge z . Potential outcomes are Y_{1i} and Y_{0i} , with

treatment effect $\delta_i = Y_{1i} - Y_{0i}$. We maintain that judge assignment is independent of $(Y_{1i}, Y_{0i}, \{D_{iz}\}_z)$ and affects outcomes only through treatment. Writing $p_z = E[D_i | Z_i = z]$ for judge z 's propensity and $\bar{p} = \sum_z \lambda_z p_z$ for the assignment-weighted average propensity, the estimand is $\beta^{IV} = E[\omega_i \delta_i] / E[\omega_i]$.

The weight on individual i is the covariance, across judge assignments, between judge propensities and i 's potential treatment profile, and its sign governs whether IV retains a weighted-average interpretation. This weight is

$$\omega_i = \sum_{z=1}^J \lambda_z (p_z - \bar{p})(D_{iz} - \bar{D}_i), \quad \bar{D}_i = \sum_{z=1}^J \lambda_z D_{iz}. \quad (2.1)$$

It is positive when judges who treat more often in the population are also more likely to treat individual i . When treatment effects are heterogeneous, an individual with $\omega_i < 0$ enters β^{IV} with negative weight, and the estimand can no longer be read as an average effect for a positively weighted subpopulation.

Strong monotonicity fixes the target population through behavior, whereas average monotonicity asks only that each individual's weight be nonnegative. Strong monotonicity requires $D_{iz} \geq D_{iz'}$ whenever $p_z \geq p_{z'}$. This pins the estimand to a fixed complier population but forces judges to differ along a single leniency dimension, a restriction that fails when judges rank cases differently across margins. Average monotonicity weakens this requirement to a sign condition on the individual weight.

Assumption 1 (Average Monotonicity) *For each individual i , $\omega_i \geq 0$ almost surely.*

Average monotonicity allows individual profiles that violate the common ranking of strong monotonicity while still preserving nonnegative weights, so β^{IV} remains a proper weighted average of treatment effects, as in [Frandsen et al. \(2023\)](#).

Average monotonicity removes negative weights but ties the admissible response types to where each judge sits relative to the average propensity, not to behavior alone. Describe individual i by a response type $\tau = (\tau_1, \dots, \tau_J) \in \{0, 1\}^J$, where τ_z is the treatment status under judge z . Because $\sum_z \lambda_z (p_z - \bar{p}) = 0$, the type-specific weight is $\omega(\tau) = \sum_z \lambda_z (p_z - \bar{p}) \tau_z$, and average monotonicity requires $\omega(\tau) \geq 0$ for every type that appears in the population. Whether a type clears this bar depends on the propensity profile $(p_1, \dots, p_J)$ and the assignment probabilities $(\lambda_1, \dots, \lambda_J)$, not on the response type alone ([Mogstad and Torgovitsky, 2024](#)). Average monotonicity therefore solves the negative-weight problem without fixing a target population from behavior alone, an observation we sharpen next.

3 Boundary Discontinuity

The three-judge design is the smallest setting in which average monotonicity changes the target population without changing the sign of the IV weights. Order the judges so that $p_1 \leq p_2 \leq p_3$. The two extreme judges generate ordinary compliers; three judges is the smallest design that also admits individuals whose treatment is non-monotone in judge propensity while leaving $\bar{p}$ free to move between them.

Written relative to the intermediate judge, the weight makes the sign of ω_i depend on treatment status under the extreme judges compared with the middle judge. Since $\sum_{z=1}^3 \lambda_z(p_z - \bar{p}) = 0$,

$$\omega_i = \lambda_1(p_1 - \bar{p})(D_{i1} - D_{i2}) + \lambda_3(p_3 - \bar{p})(D_{i3} - D_{i2}). \quad (3.1)$$

Because $p_1 \leq \bar{p} \leq p_3$, the first coefficient is nonpositive and the second nonnegative, so the sign of ω_i is set by how individual i 's treatment under the lenient and strict judges compares with treatment under the intermediate judge. Each individual belongs to one of eight response types $g = (D_{i1}, D_{i2}, D_{i3})$, enumerated with their weights and admissibility under strong and average monotonicity in Appendix A.

Six of the eight response types have an admissibility that does not depend on the propensity profile. Always-takers and never-takers receive zero weight; the monotone compliers $(0, 0, 1)$ and $(0, 1, 1)$ are admissible under both conditions; the reverse types $(1, 1, 0)$ and $(1, 0, 0)$ carry negative weight and are ruled out under average monotonicity. None of these classifications turns on where p_2 lies relative to $\bar{p}$.

The two non-monotone mirror types, $g_4 = (1, 0, 1)$ and $g_8 = (0, 1, 0)$, carry weights of opposite sign, fixed entirely by the sign of $p_2 - \bar{p}$. Type g_4 is treated by the lenient and strict judges but not the intermediate one; type g_8 is treated only by the intermediate judge. Strong monotonicity rules out both, since neither profile is weakly increasing in judge propensity. Their weights are

$$\omega(g_4) = -\lambda_2(p_2 - \bar{p}), \quad \omega(g_8) = \lambda_2(p_2 - \bar{p}). \quad (3.2)$$

When $p_2 < \bar{p}$, average monotonicity admits g_4 and excludes g_8 ; when $p_2 > \bar{p}$, the roles reverse; when $p_2 = \bar{p}$, both receive zero weight.

The admissible set therefore changes discontinuously as p_2 crosses $\bar{p}$, and when the switching types are present and have distinct average treatment effects, the estimand need not be locally invariant. All six profile-independent types keep their sign through the crossing, so the only change in the positive-weight set is the switch between g_4 and g_8 : one type exits and its mirror enters (Figure 1).

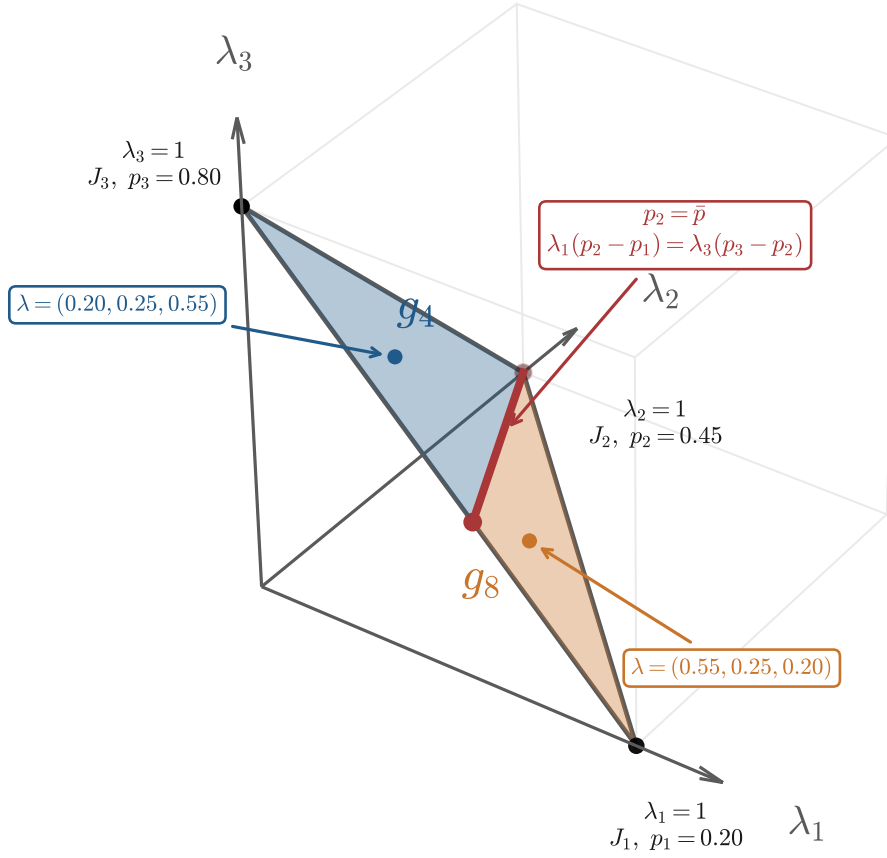


Figure 1: The Average-Monotonicity Boundary in the Assignment-Share Simplex (Three Judges)

Notes: The figure fixes judge propensities at $p_1 = 0.20$, $p_2 = 0.45$, $p_3 = 0.80$ and varies the assignment shares $(\lambda_1, \lambda_2, \lambda_3)$ over the simplex. The red locus is the average-monotonicity boundary $p_2 = \bar{p}$, equivalently $\lambda_1(p_2 - p_1) = \lambda_3(p_3 - p_2)$. In the blue region $p_2 < \bar{p}$ and the non-monotone type $g_4 = (1, 0, 1)$ receives positive weight; in the orange region $p_2 > \bar{p}$ its mirror $g_8 = (0, 1, 0)$ receives positive weight. Crossing the boundary switches the positively weighted type discretely rather than reweighting smoothly.

Proposition 1 (Boundary discontinuity) *In the three-judge design with $p_1 \leq p_2 \leq p_3$, the set of response types consistent with average monotonicity changes discontinuously across $p_2 = \bar{p}$.*

Let $\tau_g = E[\delta_i \mid \text{type } g]$. The crossing replaces a weighted average loading on g_4 with one loading on g_8 , so if the switching types carry positive mass and $\tau_{g_4} \neq \tau_{g_8}$, the identified parameter need not be locally invariant.

Corollary 1 (Target-population instability) *Across $p_2 = \bar{p}$, two average-monotonicity-consistent designs arbitrarily close in the propensity profile can place positive weight on disjoint switching types, g_4 and g_8 . If these types have positive shares in their respective populations and $\tau_{g_4} \neq \tau_{g_8}$, the resulting IV estimands target different positive-weight populations and need not be locally invariant. The discontinuity is therefore in the target population, not in the fixed-population continuity of β^{IV} as a function of the propensity profile.*

The same mechanism extends to any finite number of judges. For any judge z , the type treated only by z carries weight $\lambda_z(p_z - \bar{p})$ and its componentwise complement carries $-\lambda_z(p_z - \bar{p})$, so the two switch sign exactly when p_z crosses $\bar{p}$. The Supplementary Material (Appendix B) states and proves the finite- J result; we use the three-judge case in the text because it displays the mechanism without the additional notation. Proofs of Proposition 1 and Corollary 1 are in Appendix A.

4 A Boundary Diagnostic

Once judge propensities are estimated, the relevant question is whether the sample places each judge securely on one side of $\bar{p}$, since the exact crossing $p_z = \bar{p}$ is non-generic. Let $\hat{d}_z = \hat{p}_z - \hat{\bar{p}}$ denote judge z 's estimated distance to the boundary, with $\hat{\bar{p}} = \sum_j \hat{\lambda}_j \hat{p}_j$, and let $\hat{r}_z = |\hat{d}_z| / \widehat{\text{se}}(\hat{d}_z)$ be its standardized value. A judge with small $\hat{r}_z$ is weakly classified, and when such judges carry nontrivial assignment mass the sign of their contribution to β^{IV} , and hence the target population, is not pinned down by the data; the statistic is a sign-classification device, not a test of average monotonicity. We recommend reporting, alongside the usual balance, first-stage, and exclusion checks, each judge's $\hat{d}_z$, its standardized value $\hat{r}_z$, and the assignment share of judges near the boundary. In the Philadelphia bail data of [Stevenson \(2018\)](#), three of eight judges (about 29 percent of cases) are boundary-fragile at $c = 1.96$, even in a design earlier work does not reject ([Coulibaly et al., 2024](#)); the Supplementary Material (Appendix C) gives the construction and the full illustration, and Appendix D shows that such fragility is common in many-judge designs.

5 Conclusion

Average monotonicity gives judge-IV estimands nonnegative weights, but it does not give them a fixed target population. The positive-weight response types switch discretely when a judge’s treatment propensity crosses the assignment-weighted average, and when the switching types have positive population shares and distinct average treatment effects, that switch need not leave the IV estimand locally invariant. The estimand stays a proper weighted average; what average monotonicity does not guarantee is the population over which it is taken.

The accompanying diagnostic makes this stability checkable without testing average monotonicity itself: researchers who invoke average monotonicity should report where estimated judge propensities lie relative to the assignment-weighted average, how precisely those distances are estimated, and how much case mass sits near the boundary. When fragility is detected, a natural next step is a sensitivity analysis bounding how much the IV estimand can vary across the populations induced by alternative classifications of the fragile judges.

A Proofs for the Three-Judge Case

This appendix proves Proposition 1 and Corollary 1 and records the response-type table. Judges are ordered $p_1 \leq p_2 \leq p_3$ with $\lambda_z > 0$ and $\bar{p} = \sum_{z=1}^3 \lambda_z p_z$; a response type $g \in \{0, 1\}^3$ has type-specific weight $\omega(g) = \sum_{z=1}^3 \lambda_z (p_z - \bar{p}) g_z$.

The weight admits a representation relative to the intermediate judge. Because $\sum_{z=1}^3 \lambda_z (p_z - \bar{p}) = 0$,

$$\omega(g) = \lambda_1(p_1 - \bar{p})(g_1 - g_2) + \lambda_3(p_3 - \bar{p})(g_3 - g_2), \quad (\text{A.1})$$

where $\lambda_1(p_1 - \bar{p}) \leq 0$ and $\lambda_3(p_3 - \bar{p}) \geq 0$ because $p_1 \leq \bar{p} \leq p_3$. Evaluating (A.1) at the eight types gives the weights in Table A.1; the non-monotone mirror types satisfy $\omega(1, 0, 1) = -\lambda_2(p_2 - \bar{p})$ and $\omega(0, 1, 0) = \lambda_2(p_2 - \bar{p})$, the only weights whose sign depends on $p_2 - \bar{p}$.

Proof of Proposition 1. Since $\lambda_2 > 0$, the signs of $\omega(g_4) = -\lambda_2(p_2 - \bar{p})$ and $\omega(g_8) = \lambda_2(p_2 - \bar{p})$ are set by $p_2 - \bar{p}$. When $p_2 < \bar{p}$, $\omega(g_4) > 0$ and $\omega(g_8) < 0$, so average monotonicity admits g_4 and rules out g_8 . When $p_2 > \bar{p}$, the signs reverse, so average monotonicity admits g_8 and rules out g_4 . When $p_2 = \bar{p}$, both weights are zero. All other response types have signs that do not depend on $p_2 - \bar{p}$, so the only change in the positive-weight set as p_2 crosses $\bar{p}$ is the switch between g_4 and g_8 . The positive-weight set therefore changes discontinuously at $p_2 = \bar{p}$. ■

Table A.1: Response Types, IV Weights, and Admissibility

Type	(D_{i1}, D_{i2}, D_{i3})	$\omega(g)$	Strong monotonicity	Average monotonicity
g_1	$(1, 1, 1)$	0	Yes	Yes
g_2	$(0, 0, 0)$	0	Yes	Yes
g_3	$(0, 0, 1)$	$\lambda_3(p_3 - \bar{p}) > 0$	Yes	Yes
g_5	$(0, 1, 1)$	$-\lambda_1(p_1 - \bar{p}) > 0$	Yes	Yes
g_6	$(1, 1, 0)$	$-\lambda_3(p_3 - \bar{p}) < 0$	No	No
g_7	$(1, 0, 0)$	$\lambda_1(p_1 - \bar{p}) < 0$	No	No
g_4	$(1, 0, 1)$	$-\lambda_2(p_2 - \bar{p})$	No	Depends
g_8	$(0, 1, 0)$	$\lambda_2(p_2 - \bar{p})$	No	Depends

Notes: The table reports response types in the three-judge case, ordered by judge propensities $p_1 \leq p_2 \leq p_3$. The signs for g_3 , g_5 , g_6 , and g_7 are strict when $p_1 < \bar{p} < p_3$. At boundary cases, the corresponding weights may be zero. Under average monotonicity, a response type is admissible only if its weight is nonnegative.

Proof of Corollary 1. Write the IV estimand under average monotonicity as a ratio over positively weighted types,

$$\beta^{IV} = \frac{\sum_{g:\omega(g)>0} \pi_g \omega(g) \tau_g}{\sum_{g:\omega(g)>0} \pi_g \omega(g)},$$

where $\pi_g = \Pr(G_i = g)$ and $\tau_g = E[\delta_i \mid G_i = g]$. For a fixed population, this ratio is continuous in the propensity profile. Every weight $\omega(g)$ is continuous in p_2 , and the switching weights $\omega(g_4) = -\lambda_2(p_2 - \bar{p})$ and $\omega(g_8) = \lambda_2(p_2 - \bar{p})$ vanish as $p_2 \rightarrow \bar{p}$, so numerator and denominator are continuous and, wherever the denominator is bounded away from zero, so is β^{IV} . The admissible support is discontinuous. By Proposition 1 the admissible set switches g_4 for g_8 as p_2 crosses $\bar{p}$, since a positive share on the excluded mirror type would carry negative weight. Preserving average monotonicity across the boundary therefore requires two distinct populations, with identified parameters

$$\beta_- = \frac{\sum_{g \in \{g_3, g_4, g_5\}} \pi_g \omega(g) \tau_g}{\sum_{g \in \{g_3, g_4, g_5\}} \pi_g \omega(g)}, \quad \beta_+ = \frac{\sum_{g \in \{g_3, g_5, g_8\}} \pi_g \omega(g) \tau_g}{\sum_{g \in \{g_3, g_5, g_8\}} \pi_g \omega(g)},$$

that average over disjoint switching components and need not coincide. Example 1 gives admissible shares and heterogeneous effects with $\tau_{g_4} \neq \tau_{g_8}$ for which $\beta_- \neq \beta_+$. ■

Example 1 (Target population across the boundary) Consider two three-judge designs that differ only in whether p_2 lies just below or just above $\bar{p}$, each consistent with average monotonicity. In the first, $p_2 < \bar{p}$ and the non-monotone type present is $g_4 = (1, 0, 1)$; in the second, $p_2 > \bar{p}$ and it is the mirror $g_8 = (0, 1, 0)$. Apart from the switching type, both populations contain only always-takers and never-takers, which carry zero weight. Average monotonicity holds in each design, since the admitted non-monotone type carries positive

weight while its mirror is absent. Set $\tau_{g_4} \neq \tau_{g_8}$. The only positively weighted type is the switching type, so the first design identifies $\beta_- = \tau_{g_4}$ and the second $\beta_+ = \tau_{g_8}$. The two designs are arbitrarily close in the propensity profile yet identify effects for disjoint groups, so $\beta_- \neq \beta_+$, even though for a single fixed population β^{IV} would vary continuously in p_2 .

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